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STAT525 HOMEWORK#2

1. In four sentences or less, explain what a sampling distribution is and why it is important to know the sampling distribution of a test statistic.

2. Consider the Toluca Company example of Chapter 1. On page 46, a 95% confidence interval is constructed for the slope. Suppose instead the company were interested in the increase in mean work hours for an additional 6 units in the lot. Construct the 95% confidence interval for this quantity.

3. You fit the simple linear model to a data set and obtain estimates $b_0 = 2, b_1 = 5$, and $MSE = s^2 = 1.0$

   (a) Suppose the data consisted of $n = 20$ cases and the standard error of b_1, $s(b_1)$, is equal to 2. Construct the 95% CI for β_1.

   (b) Do you have statistical evidence X helps predict Y? Explain.

   (c) The 95% confidence interval for $E(Y)$ when $X = 5$ is [23.38, 30.62]. Find the 95% prediction interval when $X = 5$.

4. KNNL Problem 2.7

5. KNNL Problem 2.16

6. KNNL Problem 2.26. Use SAS to generate the ANOVA table and in place of part (c), generate the scatterplot and comment on the linearity.

7. Given that R^2 = SSR/SSTO, it can be shown that $R^2/(1 - R^2)$ = SSR/SSE. If you have 25 cases and an $R^2 = 0.18$, what is the F statistic for the test that the slope is equal to zero?

1. In four sentences or less, explain what a sampling distribution is and why it is important to know the sampling distribution of a test statistic.

- A sampling distribution is the probability distribution of a statistic (e.g. sample mean) computed from repeated random samples of the same size from a population.
- A sampling distribution describes how the statistic varies due to random sampling.
- Knowing the sampling distribution of a test statistic allows us to determine probabilities, critical values, and p-values under the H_0.
- This is essential for making valid statistical inferences about the population.

2. Consider the Toluca Company example of Chapter 1. On page 46, a 95% confidence interval is constructed for the slope. Suppose instead the company were interested in the increase in mean work hours for an additional 6 units in the lot. Construct the 95% confidence interval for this quantity.

For a 95 percent confidence coefficient, we require $t(.975; 23)$. From Table B.2 in Appendix B, we find $t(.975; 23) = 2.069$. The 95 percent confidence interval, by (2.15), then is:

$$3.5702 - 2.069(.3470) \leq \beta_1 \leq 3.5702 + 2.069(.3470)$$

$$2.85 \leq \beta_1 \leq 4.29$$

Thus, with confidence coefficient .95, we estimate that the mean number of work hours increases by somewhere between 2.85 and 4.29 hours for each additional unit in the lot.

TABLE 2.1 Results for Toluca Company Example Obtained in Chapter 1.

$n = 25$	$\bar{X} = 70.00$
$b_0 = 62.37$	$b_1 = 3.5702$
$\hat{Y} = 62.37 + 3.5702X$	$SSE = 54{,}825$
$\sum(X_i - \bar{X})^2 = 19{,}800$	$MSE = 2{,}384$
$\sum(Y_i - \bar{Y})^2 = 307{,}203$	

FIGURE 2.2 Portion of MINITAB Regression Output—Toluca Company Example.

```
The regression equation is
Y = 62.4 + 3.57 X

Predictor      Coef     Stdev    t-ratio        p
Constant      62.37     26.18       2.38    0.026
X            3.5702    0.3470      10.29    0.000

s = 48.82        R-sq = 82.2%        R-sq(adj) = 81.4%

Analysis of Variance

SOURCE        DF        SS        MS         F        p
Regression     1    252378    252378    105.88    0.000
Error         23     54825      2384
Total         24    307203
```

Lot size ↑ by 6 units → mean work hours ↑ by : $6 \cdot \hat{\beta}_1$

$$\boxed{CI = \hat{\beta}_1 \pm t_{\alpha/2, df} \cdot SE}$$

⇓ Additional 6 units for $\hat{\beta}_1$

① $CI = (6)\hat{\beta}_1 \pm t_{\alpha/2, df} \cdot SE\,(6)$

$CI = (6)(3.5702) \pm 2.069 \cdot 0.3470\,(6)$

$CI = 21.4212 \pm 4.3077$

- $\hat{\beta}_1 = 3.5702$
- $t_{0.975, 23} = 2.069$
- $SE = 0.3470$

$(17.1135, 25.7289) \rightarrow (17.11, 25.73)$

Lot size ↑ by 6 units

② $(6)\, 2.85 \leq \hat{\beta}_1 \leq 4.29\, (6)$

⇓

$17.1 \leq 6\hat{\beta}_1 \leq 25.74$ ✓

3. You fit the simple linear model to a data set and obtain estimates $b_0 = 2, b_1 = 5$, and $MSE = s^2 = 1.0$

(a) Suppose the data consisted of $n = 20$ cases and the standard error of b_1, $s(b_1)$, is equal to 2. Construct the 95% CI for β_1.

$\boxed{CI = \hat{\beta}_1 \pm t_{\alpha/2, df} \cdot SE}$

- $\hat{\beta}_1 = 5$
- $df = 20 - 2 = 18$
- $t_{0.975, 18} = 2.101$
- $s(b_1) = 2$

$CI = 5 \pm 2.101 \cdot 2$

$(0.798, 9.202) \rightarrow (0.80, 9.20)$

95% CI for β_1

(b) Do you have statistical evidence X helps predict Y? Explain.

① $(0.80, 9.20)$, the 95% CI for β_1, does not contain 0, meaning we reject H_0.

② $\boxed{t = \hat{\beta}_1 / SE(\hat{\beta}_1)}$ $t = 5/2 = 2.5 > t_{0.975, 18} = 2.101$, meaning we reject H_0.

Hence, there is statistical significance that the slope is different from 0.
Therefore X helps predict Y.

(c) The 95% confidence interval for $E(Y)$ when $X = 5$ is [23.38, 30.62]. Find the 95% prediction interval when $X = 5$.

$$\boxed{\begin{aligned} \text{Prediction Interval} &= \hat{Y} \pm t_{\alpha/2, df} \cdot S \cdot \sqrt{1 + h} \\ \text{Confidence Interval} &= \hat{Y} \pm t_{\alpha/2, df} \cdot S \cdot \sqrt{h} \end{aligned}}$$

- $Y = 5X + 2$
- $\hat{Y}_5 = 5(5) + 2 = 27$

$$CI \text{ Higher} = 27 + 2.101 \cdot 1 \cdot \sqrt{h} = 30.62$$
$$\sqrt{h} = 1.723$$
$$h = 2.9687 \checkmark$$

$$CI \text{ Lower} = 27 - 2.101 \cdot 1 \cdot \sqrt{h} = 23.38$$
$$\sqrt{h} = 1.723$$
$$h = 2.9687 \checkmark$$

$$PI \text{ higher} = 27 + 2.101 \cdot 1 \cdot \sqrt{1 + 2.9687} = 31.1855$$
$$PI \text{ lower} = 27 - 2.101 \cdot 1 \cdot \sqrt{1 + 2.9687} = 22.8145$$

$\Downarrow$

95% Prediction Interval when $X = 5$:
$(22.81, 31.19)$

4. KNNL Problem 2.7

2.7 Refer to **Plastic hardness** Problem 1.22.

a. Estimate the change in the mean hardness when the elapsed time increases by one hour. Use a 99 percent confidence interval. Interpret your interval estimate.

```
data hardness;
input hardness time;
datalines;
199 16
205 16
196 16
200 16
218 24
220 24
215 24
223 24
237 32
234 32
235 32
230 32
250 40
248 40
253 40
246 40
;
run;

proc reg data=hardness;
    model hardness = time / clb alpha=0.01;
run;
```

The REG Procedure
Model: MODEL1
Dependent Variable: hardness

Number of Observations Read	16
Number of Observations Used	16

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	5297.51250	5297.51250	506.51	<.0001
Error	14	146.42500	10.45893		
Corrected Total	15	5443.93750			

Root MSE	3.23403	R-Square	0.9731
Dependent Mean	225.56250	Adj R-Sq	0.9712
Coeff Var	1.43376		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	99% Confidence Limits	
Intercept	1	168.60000	2.65702	63.45	<.0001	160.69046	176.50954
time	1	2.03438	0.09039	22.51	<.0001	1.76529	2.30346

- $\hat{\beta}_1 = 2.03438$
- CI for $\hat{\beta}_1$: $(1.76529, 2.30346)$

- 2.034 units per hour in estimated change in mean hardness for every 1 hour ↑.
- With 99% confidence, we estimate that for every one-hour ↑ in elapsed time, the mean plastic hardness ↑ by between 1.77 to 2.30 units.
- CI does not contain 0, we reject H_0, there is statistical evidence that ↑ elapsed time, ↑ plastic hardness.

b. The plastic manufacturer has stated that the mean hardness should increase by 2 Brinell units per hour. Conduct a two-sided test to decide whether this standard is being satisfied; use $\alpha = .01$. State the alternatives, decision rule, and conclusion. What is the P-value of the test?

$H_0: \hat{\beta}_1 = 2$
$H_A: \hat{\beta}_1 \neq 2$

$$\boxed{t = \frac{\hat{\beta}_1 - \beta_{H_0}}{SE(\hat{\beta}_1)}}$$

$$t = \frac{2.03438 - 2}{0.09039} = 0.380352 < t_{0.995,14} = 2.624$$

p-value: 0.7071

$\Downarrow$

- Fail to reject H_0 at the $\alpha = 0.01$ significance level.
- There is no statistical evidence that the actual mean ↑ in hardness per hour is different from the manufacturer's standard of 2 Brinell units.

c. Obtain the power of your test in part (b) if the standard actually is being exceeded by .3 Brinell units per hour. Assume $\sigma\{b_1\} = .1$.

$\downarrow$ SE

- $H_0: \beta_1 = 2$
- $H_A: \beta_1 \neq 2$
- True $\beta_1 = 2.3$
- $\sigma(\hat{\beta}_1) = 0.1$
- $\alpha = 0.01$
- $df = 14$
- $t_{0.995,14} = 2.624$

$$\boxed{CI = \hat{\beta}_1 \pm t_{\alpha/2, df} \cdot SE}$$

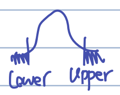
$$\boxed{Z = \frac{\text{Bound} - \beta_A}{\sigma \hat{\beta}_1}}$$

- CI higher $= 2 + 2.624 \cdot 0.1 = 2.2624$
- CI lower $= 2 - 2.624 \cdot 0.1 = 1.7376$

$(1.7376, 2.2624)$

- CI for $\hat{\beta}_1 = 2$ (Under H_0)

$P(\hat{\beta}_1 < 1.7376$ or $\hat{\beta}_1 > 2.2624)$



- Z Bound higher $= \frac{2.2624 - 2.3}{0.1} = -0.376$
- Z Bound lower $= \frac{1.7376 - 2.3}{0.1} = -5.624$

$(-5.624, -0.376)$

- Z Bound for $\beta_A = 2.3$ (Under H_A)

$\Downarrow$

$P(Z < -5.624$ or $Z > -0.376)$

$\Downarrow$

$P(Z < -5.624) \approx 0$

$P(Z > -0.376) \approx 0.64654$

- There is a 64.7% power to detect true slope of 2.3 against $H_0 = 2$ at $\alpha = 0.01$ level.

5. KNNL Problem 2.16

2.16. Refer to **Plastic hardness** Problem 1.22.

a. Obtain a 98 percent confidence interval for the mean hardness of molded items with an elapsed time of 30 hours. Interpret your confidence interval.

```
. 30
;
run;

proc reg data=hardness;
    model hardness = time / clm alpha=0.02;
    id time;
run;
```

Refer to SAS output from Q4 (2.7)

The REG Procedure
Model: MODEL1
Dependent Variable: hardness

a)

Output Statistics

Obs	time	Dependent Variable	Predicted Value	Std Error Mean Predict	98% CL Mean		Residual
1	16	199	201.1500	1.3529	197.5993	204.7007	-2.1500
2	16	205	201.1500	1.3529	197.5993	204.7007	3.8500
3	16	196	201.1500	1.3529	197.5993	204.7007	-5.1500
4	16	200	201.1500	1.3529	197.5993	204.7007	-1.1500
5	24	218	217.4250	0.8857	215.1006	219.7494	0.5750
6	24	220	217.4250	0.8857	215.1006	219.7494	2.5750
7	24	215	217.4250	0.8857	215.1006	219.7494	-2.4250
8	24	223	217.4250	0.8857	215.1006	219.7494	5.5750
9	32	237	233.7000	0.8857	231.3756	236.0244	3.3000
10	32	234	233.7000	0.8857	231.3756	236.0244	0.3000
11	32	235	233.7000	0.8857	231.3756	236.0244	1.3000
12	32	230	233.7000	0.8857	231.3756	236.0244	-3.7000
13	40	250	249.9750	1.3529	246.4243	253.5257	0.0250
14	40	248	249.9750	1.3529	246.4243	253.5257	-1.9750
15	40	253	249.9750	1.3529	246.4243	253.5257	3.0250
16	40	246	249.9750	1.3529	246.4243	253.5257	-3.9750
17	30	.	229.6313	0.8285	227.4569	231.8056	.

Sum of Residuals	0
Sum of Squared Residuals	146.42500
Predicted Residual SS (PRESS)	194.01315

$(227.4569, 231.8056)$

- With 98% confidence, the mean hardness of molded items with an elapsed time of $X = 30$ hours is between 227.46 to 231.81 Brinell units.

$\hat{Y} = 2.03438 X + 168.60$

$\hat{Y}_{30} = 2.03438(30) + 168.60 = 229.631$ ✓

b. Obtain a 98 percent prediction interval for the hardness of a newly molded test item with an elapsed time of 30 hours.

```
proc reg data=hardness;
    model hardness = time / cli alpha=0.02;
    id time;
run;
```

The REG Procedure
Model: MODEL1
Dependent Variable: hardness

Output Statistics

Obs	time	Dependent Variable	Predicted Value	Std Error Mean Predict	98% CL Predict		Residual
1	16	199	201.1500	1.3529	191.9496	210.3504	-2.1500
2	16	205	201.1500	1.3529	191.9496	210.3504	3.8500
3	16	196	201.1500	1.3529	191.9496	210.3504	-5.1500
4	16	200	201.1500	1.3529	191.9496	210.3504	-1.1500
5	24	218	217.4250	0.8857	208.6248	226.2252	0.5750
6	24	220	217.4250	0.8857	208.6248	226.2252	2.5750
7	24	215	217.4250	0.8857	208.6248	226.2252	-2.4250
8	24	223	217.4250	0.8857	208.6248	226.2252	5.5750
9	32	237	233.7000	0.8857	224.8998	242.5002	3.3000
10	32	234	233.7000	0.8857	224.8998	242.5002	0.3000
11	32	235	233.7000	0.8857	224.8998	242.5002	1.3000
12	32	230	233.7000	0.8857	224.8998	242.5002	-3.7000
13	40	250	249.9750	1.3529	240.7746	259.1754	0.0250
14	40	248	249.9750	1.3529	240.7746	259.1754	-1.9750
15	40	253	249.9750	1.3529	240.7746	259.1754	3.0250
16	40	246	249.9750	1.3529	240.7746	259.1754	-3.9750
17	30	.	229.6313	0.8285	220.8695	238.3930	.

(220.8695, 238.3930)

- With 98% confidence, the new hardness of molded items with an elapsed time of $X = 30$ hours is between 220.87 to 238.39 Brinell units.

c. Obtain a 98 percent prediction interval for the mean hardness of 10 newly molded test items, each with an elapsed time of 30 hours.

$$\boxed{SE_{adj} = \sqrt{SE + (MSE/m)}}$$

$$\boxed{\text{Prediction Interval} = \hat{Y} \pm t_{\alpha/2,\,df} \cdot SE_{adj}}$$

Refer to SAS output from Q4 (2.7)

- $MSE = 10.45893$
- $m = 10$
- $SE(\hat{Y}_{30}) = 0.8285$
- $t_{0.01,14} = 2.6245$

$$SE_{adj} = \sqrt{(0.8285)^2 + (10.45893/10)} = 1.31617$$

$$\text{Prediction Interval} = 229.631 \pm 2.6245(1.31617)$$

$$\Downarrow$$

$$(226.177, 233.085)$$

- With 98% confidence, the 10 new hardness of molded items with an elapsed time of $X = 30$ hours is between 266.18 to 233.09 Brinell units.

d. Is the prediction interval in part (c) narrower than the one in part (b)? Should it be?

Yes. Interval in (c) is narrower in (d), as it should be.

- Averaging over 10 items ↓ variability, ↓SE, hence ↓ interval.

WH Band

e. Determine the boundary values of the 98 percent confidence band for the regression line when $X_h = 30$. Is your confidence band wider at this point than the confidence interval in part (a)? Should it be?

```
proc reg data=hardness;
    model hardness = time / clm cli alpha=0.02;  /* 98% CI and PI */
    output out=a2 p=predicted stdp=stdp uclm=uclm lclm=lclm ucl=ucl lcl=lcl;
    id time;
run;

data a3;
    set a2;
    /* df = n - 2 = 16 - 2 = 14 */
    whl = predicted - sqrt(2 * finv(1 - 0.02, 2, 14)) * stdp;
    whu = predicted + sqrt(2 * finv(1 - 0.02, 2, 14)) * stdp;
run;

proc print data=a3;
    where time = 30;
    var time predicted whl whu lclm uclm lcl ucl;
run;
```

Obs	time	predicted	whl	whu	lclm	uclm	lcl	ucl
17	30	229.631	226.949	232.313	227.457	231.806	220.869	238.393

(a): CI　　(b): PI

(226.95, 232.31)
Working-Hotelling Band
Confidence Band

- Yes. WH Band is slightly wider than regular CI for the mean, as it should be.
- WH accounts for simultaneous inference across all values of X, not just at X=30.
- WH is more conservative than CI, which is why it is wider.

6. KNNL Problem 2.26. Use SAS to generate the ANOVA table and in place of part (c), generate the scatterplot and comment on the linearity.

2.26. Refer to **Plastic hardness** Problem 1.22.

a. Set up the ANOVA table.

Refer to SAS output from Q4 (2.7)

```
proc reg data=hardness;
    model hardness = time;
run;
```

The REG Procedure
Model: MODEL1
Dependent Variable: hardness

Number of Observations Read	16
Number of Observations Used	16

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	5297.51250	5297.51250	506.51	<.0001
Error	14	146.42500	10.45893		
Corrected Total	15	5443.93750			

b. Test by means of an F test whether or not there is a linear association between the hardness of the plastic and the elapsed time. Use $\alpha = .01$. State the alternatives, decision rule, and conclusion.

Refer to SAS output from Q4 (2.7)

$H_0: B_1 = 0$ (No Linear Trend)
$H_A: B_1 \neq 0$ (Linear Trend)

```
proc reg data=hardness;
    model hardness = time / alpha=0.01;
run;
```

- F-statistic : 506.51
- p-value : < 0.0001 < 0.01 = α

$\Downarrow$

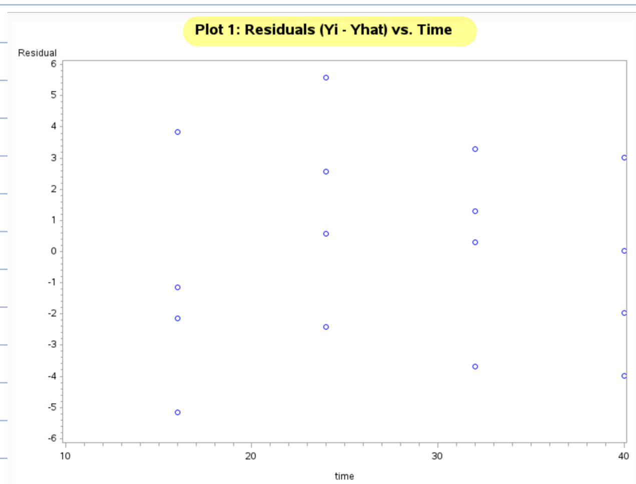
Reject H_0

- There is strong evidence of linear relationship between elapsed time and plastic hardness.
- Time is a statistically significant predictor of hardness at 1% level.

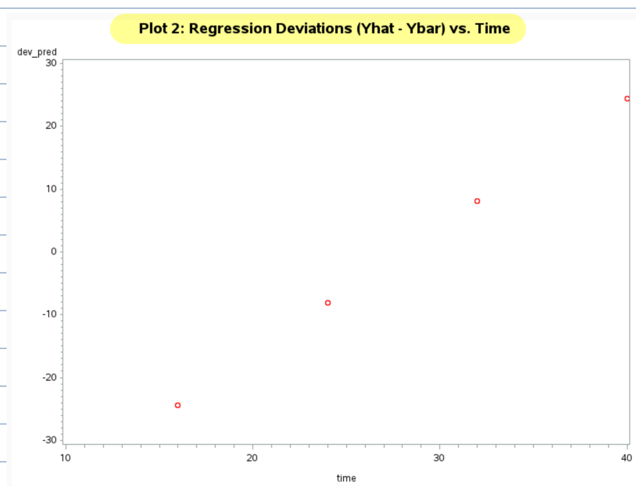
c. Plot the deviations $Y_i - \hat{Y}_i$ against X_i on a graph. Plot the deviations $\hat{Y}_i - \bar{Y}$ against X_i on another graph, using the same scales as for the first graph. From your two graphs, does SSE or SSR appear to be the larger component of $SSTO$? What does this imply about the magnitude of R^2?

$\boxed{\begin{array}{l} e = Y_i - \hat{Y} = \text{Residuals} \\ \hat{Y}_i - \bar{Y} = \text{Deviation from the mean} \end{array}}$

```
proc reg data=hardness;
    model hardness = time;
    output out=regout p=pred r=resid; /* Save Yhat and residuals */
run;

proc means data=regout mean noprint;
    var hardness;
    output out=meanout mean=Ybar;
run;

data regout2;
    if _n_ = 1 then set meanout;
    set regout;
    dev_pred = pred - Ybar; /* SSR component */
run;

/* Plot Residuals vs Time (Yi - Yhat) */
title "Plot 1: Residuals (Yi - Yhat) vs. Time";
symbol1 v=circle c=blue i=none;
proc gplot data=regout2;
    plot resid*time;
    axis1 label=("Elapsed Time");
    axis2 label=("Residual");
run;
quit;

/* Plot Deviations (Yhat - Ybar) vs Time */
title "Plot 2: Regression Deviations (Yhat - Ybar) vs. Time";
symbol1 v=circle c=red i=none;
proc gplot data=regout2;
    plot dev_pred*time;
    axis1 label=("Elapsed Time");
    axis2 label=("Deviation from Mean");
run;
```



random scatter → good.



linear trend → good.

$SSR = 5297.5125 \ggg 146.4250 = SSE$

- SSR is much larger than SSE

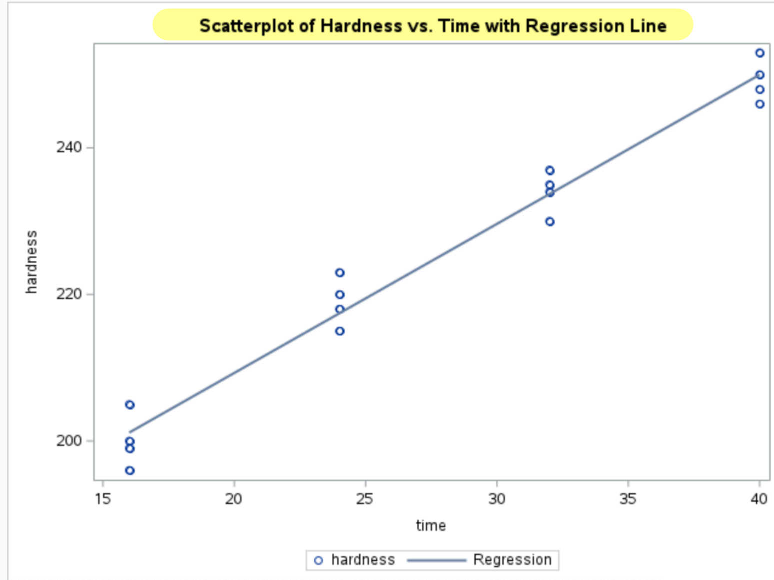
$R^2 = 5297.5125 / 5443.9375 = 0.9731$

- This high value (97.31%) confirms that most of total variation, (SSTO), is explained by the regression (SSR), not by residuals (SSE).

↓ Linearity Assumption is well satisfied ↑

```
proc sgplot data=hardness;
    scatter x=time y=hardness;
    reg x=time y=hardness; /* Adds fitted regression line */
    title "Scatterplot of Hardness vs. Time with Regression Line";
run;
```

- $\hat{Y}$ v.s. X shows points follow a straight-line pattern along the fitted regression line with no curvature nor systematic deviation.
- Linear model is appropriate.



linear trend → good.

d. Calculate R^2 and r.

$\boxed{R^2 = SSR/SSTO}$ $\qquad$ $r = \sqrt{R^2}$

$R^2 = 5297.5125 / 5443.9375$ $\qquad$ $r = \sqrt{0.9731}$

$R^2 = 0.9731$ $\qquad$ $r = 0.9865$

- Since $\hat{\beta}_1$ is positive, correlation, r, is also positive.

7. Given that $R^2 = \text{SSR/SSTO}$, it can be shown that $\boxed{R^2/(1-R^2) = \text{SSR/SSE}.}$ If you have 25 cases and an $R^2 = 0.18$, what is the F statistic for the test that the slope is equal to zero?

$$\boxed{F = \frac{MSR}{MSE} = \frac{SSR/1}{SSE/(n-2)}}$$

$$\boxed{F = \frac{SSR(n-2)}{SSE}}$$

⇓ plug in $R^2/(1-R^2) = SSR/SSE$

$$F = (R^2/1-R^2) \cdot (n-2)$$

$$F = (0.18/1-0.18) \cdot (25-2) = 5.04878$$